Count 함수

```
count = 0 ; // 일단 count = 0
a = 0000 & 0001 = 0000 (참) ... count = 1
a = 0000 & 0010 = 0000 (참) ... count = 2
a = 0000 & 0100 = 0000 (참) ... count = 3
a = 0000 & 1000 = 0000 (참) ... count = 4 ... stop
```

Power 함수

```
a = 16 ; // 10000
b = 5 ; // 00101
result = 1 ; // 00001
while (b != 0) { // b != 0 ... count = 0 (false)
}
a = 16 ; // 10000
b = 5 ; // 00101
while (b != 0) { // b != 0 ... b = b >> 1
```

연산 결과를 보관

```
a = 16 ; // 10000
b = 5 ; // 00101
c = a & b // 10000 & 00101 = 00000 ... 16
```

Toggle 함수들

```
a = 16 ; // 00010000 (binary)
toggle(a, 3) // 00010000 ^ (1 << 3) = 00010000 ^ 00001000 = 00011000
toggle(a, 4) // 00011000 ^ (1 << 4) = 00011000 ^ 00010000 = 00001000 ... 8
```

Change 함수들

```
a = 16 ; // 00010000 (binary)
change(a, 3, 1) // 00010000 | (1 << 3) = 00010000 | 00001000 = 00011000
change(a, 4, 0) // 00011000 & ~(1 << 4) = 00011000 & 11110111 = 00010000
change(a, 3, 0) // 00010000 & ~(1 << 3) = 00010000 & 11110111 = 00010000
change(a, 4, 1) // 00010000 | (1 << 4) = 00010000 | 00010000 = 00010000
```

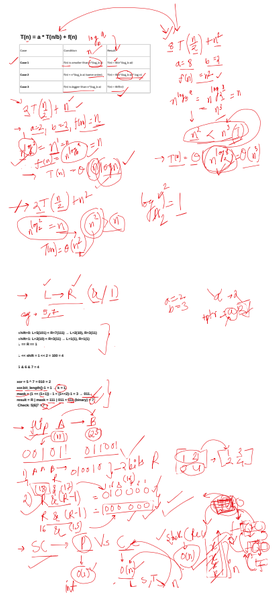
Mask

```
a = 16 ; // 00010000
mask = 8 ; // 00001000
mask & a(mask) = 00001000 & 00010000 = 00000000 ... 0
```

```
a = 16 ; // 00010000
mask = 8 ; // 00001000 (binary) ... 8
a ^ mask = 00010000 ^ 00001000 = 00011000
```

Exactly specific 비교

```
a = 16 ; // 00010000
mask = 8 ; // 00001000 ... 8
mask << (sizeof(a) - 1) ... 8 << 7 ... 1024
mask << (sizeof(a) - 1) & a = 1024 & 16 = 0000
Apply mask: 00010 & 1024 = 0000 ... 0
```

Scale 함수 비교 for range

```
a = 16 ; // 00010
b = 5 ; // 00101
0.0 <= a <= 255 ... 0 <= 16 <= 255 ... 0 <= a ... 00000
b <= 255 ... b <= 255 ... b <= 255 ... 00000
b <= mask << (sizeof(a) & 1024) << (sizeof(a) & 1024) ... 255
```